

Work Session Documentation: Telemetry, isoSPI, and Fault Verification

Subsystems: Master Board Firmware, isoSPI Link, BMS Viewer | Equipment: ST-LINK, serial terminal, oscilloscope

Session Objective

The objective of this work session was to verify that the master board could execute firmware, communicate with the LTC6811 over isoSPI, transmit UART telemetry, and assert the BMS fault output during unsafe or invalid conditions.

Raw Data and Observations

Test	Observed Result	Immediate Inference
STM32 flashing	Firmware programmed successfully using ST-LINK and STM32CubeIDE.	MCU bring-up path was functional.
UART startup	Startup output observed at 115200 baud.	Clock and UART configuration were correct.
Telemetry stream	Stable telemetry observed at 10 Hz with valid CRC.	Packet framing and CRC generation were working.
isoSPI read cycles	500 read cycles completed with 0 CRC errors.	LTC6820/LTC6811 communication path was reliable during test.
Fault output	Fault line asserted low within 20 ms during injected fault cases.	Firmware threshold logic and GPIO fault output were working.
BMS Viewer	Viewer updated at approximately 1.1 Hz and rejected invalid packets.	Dashboard validation prevented corrupted data display.

What Was Tried

Firmware was flashed to the STM32 and basic execution was checked using LED heartbeat and UART startup text. The UART stream was connected to a serial terminal and then to the BMS Viewer. LTC6811 conversions were triggered over isoSPI and repeated read cycles were monitored for CRC failures. Fault conditions were injected or simulated to observe both the hardware fault line and viewer fault indicators.

What Happened

The firmware executed correctly and telemetry packets were transmitted at 10 Hz. The isoSPI communication test completed 500 read cycles with 0 CRC errors, showing that the LTC6820 and LTC6811 link was stable during the test. The BMS fault output asserted low within 20 ms for tested fault cases, while the viewer displayed fault indicators within the expected human-interface timing range.

Inference

The session showed that the communication design was robust enough for prototype verification. The main technical inference was that embedded safety data needs multiple layers of validation: reliable isoSPI transfer from the monitor IC, CRC-protected UART telemetry to the viewer, and fast local fault assertion that does not depend on the user interface refresh rate.